



Environmental Science

Year 1

Science Department



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Topic 1: What is an Ecosystem?

Living and non-living factors:

- Living organisms share the following seven characteristics:

movement, reproduction, response to stimuli, taking in nutrients, excreting waste, respiring, and growth.

- Non-living things cannot carry out all seven of these processes.



Activity 1:

Circle the living organisms:



Living and non-living things are both important parts of an **ecosystem**.

Ecosystems:

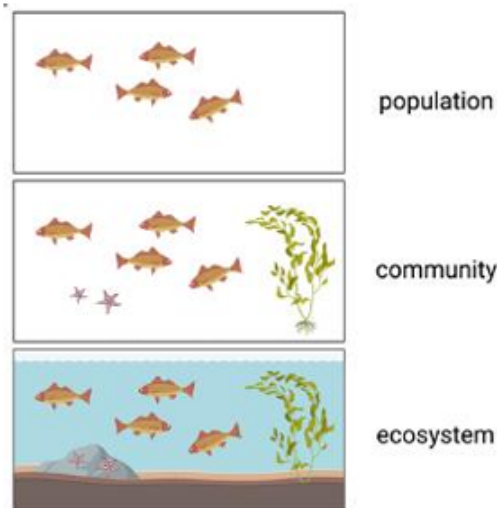
An **ecosystem** is a community of living organisms interacting with each other and with their physical environment.

It includes:

- **Living things (biotic factors):** Plants, animals, fungi, microorganisms.
- **Non-living things (abiotic factors):** Air, water, soil, sunlight, temperature.

Community: All organisms of **different species** which live in a particular area, for example all the plants, animals, bacteria and fungi in a forest.

Population: All organisms of **the same species** which live in a particular area, for example all of the cats living in Larnaca.



Living things, like plants and animals, depend on non-living things such as water, air, sunlight, and soil to survive.

- For example, plants need sunlight and water to grow, and animals need air to breathe and water to drink. Without these non-living parts, living things could not live.



Activity 2:

What is the difference between a population and a community?

Circle one correct statement.

- A population is all the different living things in one area, and a community is one type of living thing.
- A population is one type of living thing in an area, and a community is all the different living things living together.
- A population and a community mean the same thing.
- A community is only made of animals, not plants.

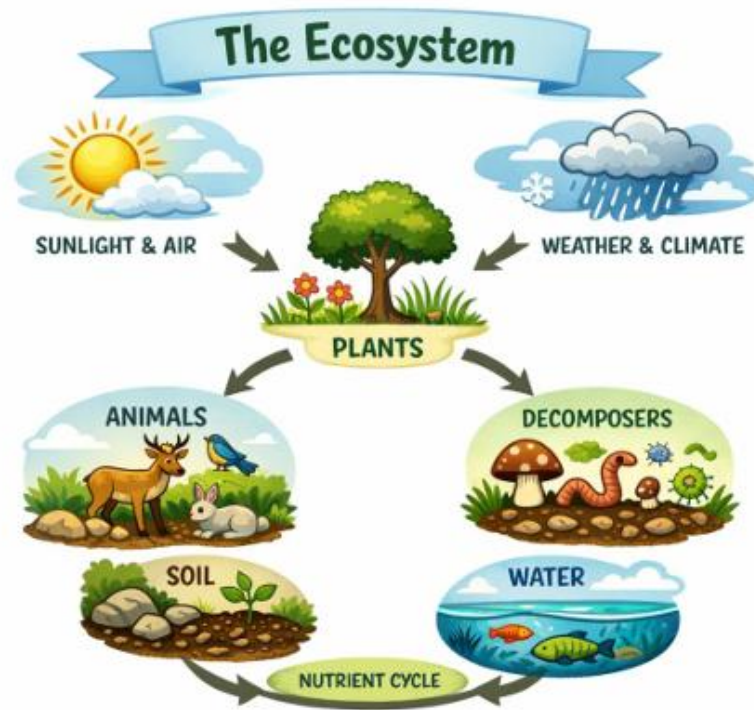
Can you give an example of a population and a community that might be in a pond?

Population:

Community:

All of these work together as a system! Everything in an ecosystem is **connected**!

A change in one part (like pollution or loss of a species) can affect the whole system.



Let's see an example:

A forest ecosystem includes:

- Trees, birds, insects, and animals
- Soil, water, sunlight, and climate



The plants produce food using sunlight, animals eat the plants (or other animals), and decomposers break down dead material—recycling nutrients back into the system.

Main types of ecosystems:

1. Terrestrial (land) Ecosystems - These are ecosystems found on land:

- a) **Forests:** Lots of trees, animals, and rainfall (e.g., rainforests).
- b) **Deserts:** Very dry, with plants and animals adapted to little water.
- c) **Grasslands:** Open areas with grasses and few trees (home to grazing animals).
- d) **Tundra:** Very cold, with frozen ground and limited plant life.

2. Aquatic Ecosystems - These exist in water and are divided into two main types:

a) Freshwater Ecosystems - Because of their low salt content, they serve as home to fish, frogs, and freshwater plants.

- a. Rivers
- b. Lakes
- c. Ponds

b) Marine Ecosystems - They have a high salt content, and they cover most of Earth.

- a. Oceans
- b. Coral reefs
- c. Seas

3. Artificial Ecosystems - Created or controlled by humans. They need human care to survive.

- a. Farms
- b. Gardens
- c. Aquariums